

# Jingyu (Jack) Zhang

Department of Computer Science · Johns Hopkins University · Baltimore, MD

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## RESEARCH INTERESTS

My research interest lies in the area of natural language processing. I am particularly interested in the **responsible development and deployment of foundation models**. My recent work focuses on safety alignment and enhancing attribution of LLMs.

## EDUCATION

**Johns Hopkins University**

Aug 2023 - Present

*Ph.D. in Computer Science*

Advisors: Daniel Khashabi and Benjamin Van Durme

**Johns Hopkins University**

Aug 2019 - May 2023

*B.S. in Computer Science*

GPA: 3.97/4.00

Additional Majors: Mathematics; Applied Mathematics & Statistics. Minor: Economics

## PUBLICATIONS & PREPRINTS

### Selected Works.....

1. **Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements**

**Jingyu Zhang**, Ahmed Elgohary, Ahmed Magooda, Daniel Khashabi, Benjamin Van Durme.

arXiv preprint: <https://arxiv.org/abs/2410.08968>

2. **Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data**

**Jingyu Zhang**, Marc Marone, Tianjian Li, Benjamin Van Durme, Daniel Khashabi.

arXiv preprint: <https://arxiv.org/abs/2404.03862>

3. **SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation**

Abe Bohan Hou\*, **Jingyu Zhang**\*, Tianxing He\*, Yichen Wang, Yung-Sung Chuang, Hongwei Wang, Lingfeng Shen, Benjamin Van Durme, Daniel Khashabi, Yulia Tsvetkov.

NAACL 2024. [\*Equal Contribution]

4. **On the Blind Spots of Model-Based Evaluation Metrics for Text Generation**

Tianxing He\*, **Jingyu Zhang**\*, Tianle Wang, Sachin Kumar, Kyunghyun Cho, James Glass, Yulia Tsvetkov.

In *Proc. of ACL 2023*. **Oral Presentation**. [\*Equal Contribution]

### All Publications.....

1. **Rationalyst: Pre-training Process-Supervision for Improving Reasoning**

Dongwei Jiang, Guoxuan Wang, Yining Lu, Andrew Wang, **Jingyu Zhang**, Chuyu Liu, Benjamin Van Durme, Daniel Khashabi.

arXiv preprint: <https://arxiv.org/abs/2410.01044>

2. **Core: Robust Factual Precision Scoring with Informative Sub-Claim Identification**

Zhengping Jiang, **Jingyu Zhang**, Nathaniel Weir, Seth Ebner, Miriam Wanner, Kate Sanders, Daniel Khashabi, Anqi Liu and Benjamin Van Durme.

arXiv preprint: <https://arxiv.org/abs/2407.03572>

3. **Self-(In)Correct: LLMs Struggle with Refining Self-Generated Responses.**

Dongwei Jiang, **Jingyu Zhang**, Orion Weller, Nathaniel Weir, Benjamin Van Durme, Daniel Khashabi.

arXiv preprint: <https://arxiv.org/abs/2404.04298>

4. **TurkingBench: A Challenge Benchmark for Web Agents**

Kevin Xu, Yeganeh Kordi, Kate Sanders, Yizhong Wang, Adam Byerly, **Jingyu Zhang**, Benjamin Van Durme, Daniel Khashabi.

arXiv preprint: <https://arxiv.org/abs/2403.11905>

5. **DiffNorm: Self-Supervised Normalization for Non-autoregressive Speech-to-speech Translation.**

Weiting Tan, **Jingyu Zhang**, Lingfeng Shen, Daniel Khashabi, Philipp Koehn.

NeurIPS 2024.

6. **k-SemStamp: A Clustering-Based Semantic Watermark for Detection of Machine-Generated Text**

Abe Bohan Hou, **Jingyu Zhang**, Yichen Wang, Daniel Khashabi, Tianxing He.  
*Findings of ACL 2024.*

7. **The Language Barrier: Dissecting Safety Challenges of LLMs in Multilingual Contexts**  
Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, **Jingyu Zhang**, Haoran Xu, Boyuan Zheng, Philipp Koehn, Daniel Khashabi.  
*Findings of ACL 2024.*
8. **On the Zero-Shot Generalization of Machine-Generated Text Detectors**  
Xiao Pu, **Jingyu Zhang**, Xiaochuang Han, Yulia Tsvetkov, Tianxing He.  
*Findings of EMNLP 2023.*
9. **Geo-Seq2seq: Twitter User Geolocation on Noisy Data through Sequence to Sequence Learning**  
**Jingyu Zhang**, Alexandra DeLucia, Chenyu Zhang, Mark Dredze.  
*Findings of ACL 2023.*
10. **PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation**  
**Jingyu Zhang**, James Glass, Tianxing He.  
\*SEM 2023. Preliminary version accepted at *2nd Workshop on Efficient Natural Language and Speech Processing (ENLSP)*, NeurIPS 2022. **Best Paper Award.**
11. **Changes in Tweet Geolocation over Time: A Study with Carmen 2.0**  
**Jingyu Zhang**, Alexandra DeLucia, Mark Dredze.  
*8th Workshop on Noisy User-generated Text (W-NUT)*, COLING 2022.
12. **Study of Manifestation of Civil Unrest on Twitter**  
Abhinav Chinta\*, **Jingyu Zhang**\*, Alexandra DeLucia, Anna L. Buzcak, Mark Dredze.  
*7th Workshop on Noisy User-generated Text (W-NUT)*, EMNLP 2021. [\*Equal Contribution]

## RESEARCH EXPERIENCE

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### Microsoft Responsible AI Research

May 2024 - Aug 2024, P/T Sep 2024 - Present

Research Intern

Mentor: Ahmed Elgohary Ghoneim

### Center for Language and Speech Processing at Johns Hopkins University

Fall 2023 - Present

PhD Researcher

Advisors: Daniel Khashabi and Benjamin Van Durme

### University of Washington

Summer 2022 - Spring 2023

Undergraduate Research Intern

Advisors: Yulia Tsvetkov and Tianxing He

### MIT Computer Science and Artificial Intelligence Laboratory

Spring 2022 - Summer 2022

Undergraduate Research Intern

Advisors: James Glass and Tianxing He

### Center for Language and Speech Processing at Johns Hopkins University

Spring 2021 - May 2023

Undergraduate Researcher

Advisors: Mark Dredze and Benjamin Van Durme

## AWARDS & HONORS

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- **Best Paper Award** - ENLSP Workshop at NeurIPS 2022
- **Michael J. Muuss Research Award** - 1 out of 744 JHU CS undergraduates (2023). Press coverage: [🔗](#)
- **CRA Outstanding Undergraduate Researcher Award Nominee** - 4 out of 744 JHU CS undergraduates (2022)
- **Pistrutto Research Fellowship** - \$4000 grant (Fall 2022). Press coverage: [🔗](#)
- **Bloomberg Distinguished Professor (BDP) Summer Program Recipient** - \$6000 grant (Summer 2021).
- **Upsilon Pi Epsilon** - International Honor Society for the Computing and Information Disciplines.
- **National Olympiad in Informatics in Provinces (NOIP)** - National 1st Prize Certification (2018)

## TALKS

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### Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements

Invited talk at CLSP Seminar, Johns Hopkins University, Baltimore, Maryland

Oct 2024

### SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation

Invited talk at CLSP Seminar, Johns Hopkins University, Baltimore, Maryland

Dec 2023

### On the Blind Spots of Model-Based Evaluation Metrics for Text Generation

Oral Presentation, ACL 2023, Toronto, Canada

July 2023

## PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation

Spotlight Presentation, ENLSP Workshop, NeurIPS 2022, New Orleans, USA

Dec 2022

## INDUSTRY EXPERIENCE

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### Microsoft

Responsible AI Research

### Research Intern

May 2024 - Present (P/T after Aug 2024)

- Conducting research related to safety alignment and automatic red teaming
- Resulting manuscript submitted to a top-tier machine learning conference, with pre-print available at <https://arxiv.org/abs/2410.08968>.

### ByteDance Ltd.

Lark Explorer Department

### C++ Development Intern

May 2020 - May 2021 (P/T after Aug 2020)

- Carried out C++ cross-platform development interacting with macOS kernel and Windows Win32 API
- Performed client-side development with Electron and Node.js related to performance optimization
- Conducted data science analytics on extensive user-generated data with Apache Hive and Python

## TEACHING EXPERIENCE

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### EN.601.465/665 Natural Language Processing

JHU Department of Computer Science

### Course Assistant

Fall 2021, Fall 2022

- EN.601.465/665 is a mixed graduate / upper-level undergraduate course in NLP taught by Prof. Jason Eisner
- Conducted grading of homework and exam papers and held review sessions and office hours on a weekly basis
- Received an average score of 4.80/5.00 on student TA evaluation (100% “Good” or “Excellent” rating)

### Code in Place 2021

Stanford University Department of Computer Science

### Section Leader (Volunteer)

April 2021 - May 2021

- Worked with a team of more than 50 teaching leads and 1000 section leaders to support 10,000+ students across the world as they navigate the first five weeks of CS 106A: Programming Methodology course
- Prepared materials and taught a Python programming section of 10 students on a weekly basis

## ACADEMIC SERVICE

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- Application Mentor, JHU CLSP pre-application support program (2023-present)
- Curriculum Committee, Department of Computer Science, Johns Hopkins University (2023-present)
- Recruitment Committee, Center for Language and Speech Processing, Johns Hopkins University (2023-present)

## SKILLS & COURSEWORK

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### Technical Skills

- Programming Languages: Python, C/C++, Java, OCaml, HTML, CSS, Javascript, MATLAB, SQL
- Frameworks: Huggingface, PyTorch, Sklearn, Pandas, Numpy, Electron, Windows/macOS native APIs
- Development Workflow: L<sup>A</sup>T<sub>E</sub>X, Bash, Emacs, Git, Makefile, GN build

### Related Coursework

- NLP/ML: Natural Language Processing, Machine Translation, Multilingual NLP, Artificial Agents, Machine Learning, Deep Learning, Human-Computer Interaction, Probabilistic Models of the Visual Cortex
- Math/Stats: Real Analysis, Abstract Algebra, Topology, Differential Equations, Probability, Statistics, Optimization, Time Series Analysis, Game Theory, Mathematical & Computational Foundations of Data Science